



Risk-Adaptive Neurosymbolic Intent Planning for Optical Networks

A Pre-Deployment Decision Mechanism with Joint Semantic and QoT Assessment

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OUTLINE

[01] Bottleneck

The Optical Intent Planning Bottleneck

Physical-layer constraints, token saturation, and hallucinated routing in optical backbones

[02] Architecture

Neurosymbolic Intent Planning Pipeline

Decoupling probabilistic reasoning (NL to PDDL) from deterministic solvers and physics tools

[03] Risk Gates

Pre-Deployment Risk Gates: Semantic & Physical Validation

Sequential fail-fast decision via Layer 1/2 semantic gate (U_sem) and GN-model QoT gate

[04] Evaluation

Experimental Testbed Validation on 17-Node Optical Topology

Benchmarking safety, human intervention reduction, and orchestration latency against baselines

[05] Outlook

Key Takeaways, System Guarantees & Future Directions

Summary of thesis contributions, operational guarantees, and extension to joint compute scheduling

Motivation: The Vision of Intent-Based Optical Networks

⚠ Operational Shift: Manual Bottleneck

- Optical backbones carry terabits of core traffic across ROADM networks
- Traditional workflow: manual CLI scripts and complex RESTConf payloads
- Human configuration delays lightpath provisioning by hours or days
- Prone to fatal human operator errors across multi-vendor optical links
- Goal: Transition to autonomous Intent-Based Networking (IBN)

⚡ Operational Promise: Autonomous Vision

- High-level abstraction: specify WHAT is needed, not HOW to configure it
- Example: 'Establish a 400G lightpath between Milan and Rome avoiding L2'
- Autonomous translation into validated physical lightpaths in seconds
- Reduces human configuration error across multi-vendor optical links
- Critical challenge: Optical networks do not tolerate probabilistic errors

Operational Flow: [Operator NL Intent] → [AI Intent Orchestrator] → [Zero-Error Physical Lightpath]

Illustrative Failure: Why Standard LLMs Break Optical Backbones

⚠ Challenge 1: Token Budget Saturation

- Full optical topology payloads (RESTConf JSON) exceed LLM context budgets
- In a 100-node core network, telemetry dumps consume tens of thousands of tokens
- Induces severe 'lost-in-the-middle' attention degradation
- Result: The LLM drops explicit user constraints such as link exclusion rules
- High API token cost and unpredictable prompt execution times

🚫 Challenge 2: Hallucinated Physics

- LLMs are probabilistic text predictors, not optical physics calculators
- Incapable of computing Generalized Signal-to-Noise Ratio (GSNR)
- Ignore nonlinear fiber Kerr effects and EDFA noise accumulation
- Result: Proposes lightpaths with unfeasible optical Quality of Transmission
- Causes severe traffic drop or optical controller rejection upon deployment

Empirical Risk: Unconstrained LLMs allow up to 34% physically unfeasible routes to reach the optical controller

Problem Statement: Inputs, Constraints & Objectives



1. Given Inputs

- Unstructured Natural Language intent from operator
- Physical topology graph $G(V, E)$ via RESTConf
- Link parameters: span lengths, attenuation, dispersion
- EDFA amplifier gains and noise figures
- Transponder specs: baud rates, modulation formats



2. Constraints

- Semantic alignment: formal model matches intent
- Optical GSNR exceeds transponder threshold
- Receiver power satisfies sensitivity bounds
- Zero spectral overlap and wavelength collision

Formulation:

$$\begin{aligned}U_{sem} &\leq \tau_{sem} \\ GSNR &\geq GSNR_{th} \\ P_{rx} &\geq P_{rx,min}\end{aligned}$$



3. Objectives

- Zero unfeasible routes reaching network controller
- Fail-fast pre-deployment validation pipeline
- Selective, risk-proportional HITL engagement
- Sub-second deterministic computation time
- Auditable planning report for network engineers

Proposed Solution: Neurosymbolic Decoupling

Probabilistic Reasoning Layer

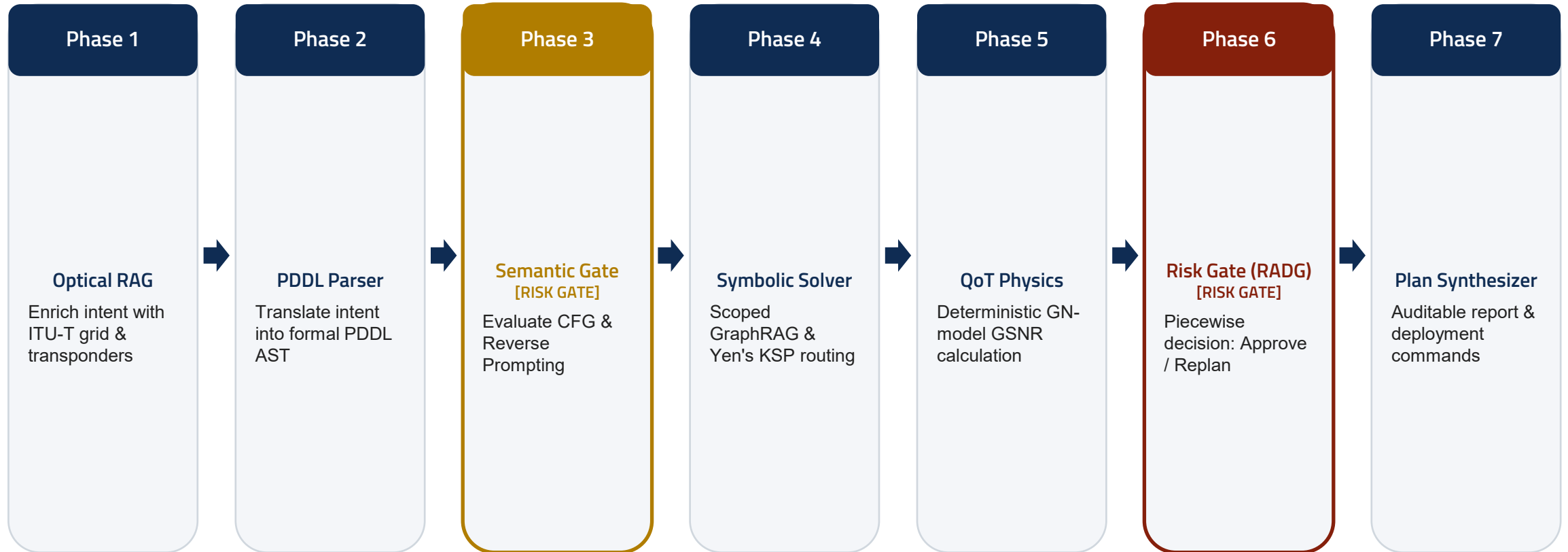
- LLMs Reason, Deterministic Tools Calculate
- Prohibit LLMs from performing graph routing or arithmetic
- Constrain the LLM strictly to formal linguistic translation
- Natural Language is parsed into formal PDDL constraints
- Zero hallucination of routing decisions or optical physics

Deterministic Execution Layer

- Path computation delegated to Yen's KSP graph algorithms
- Physical validation delegated to analytical GN-model engine
- Subtopology extraction via scoped Mock GraphRAG
- Pre-deployment safety gate (RADG) verifies feasibility
- Guarantees provable physical safety before configuration push

Core Contributions: [1] Neurosymbolic Pipeline | [2] Scoped GraphRAG | [3] Risk-Adaptive Gates | [4] LangGraph Orchestrator

End-to-End System Architecture & Pipeline Flow



⏸ Phase 3b: Clarify Loop (LangGraph interrupt())

If $U_{sem} > \tau_{sem}$: Pauses execution and prompts operator for intent disambiguation

🔄 Phase 6: Suggest Replan Loop (Physics Unfeasible)

If $QoT_{valid} = 0$: Suggests operator to relax constraints (lower baud rate, alternate link)

Overcoming Token Saturation: Scoped Optical GraphRAG

Deterministic k-hop Subtopology Scoping

- Full topology dumps overwhelm LLM context windows
- Raw JSON contains excessive telemetry: ROADM ports, EDFAs, fibers
- Mock GraphRAG extracts only the k-hop neighborhood between endpoints
- Filters out irrelevant core subnets, links, and unused transponders
- Quantitative Impact: Over 75% reduction in prompt token payload
- Sub-millisecond graph extraction: $O(V + E)$ executed in pure Python
- Guarantees sharp LLM attention focus on active optical constraints



Visual Diagram: Full 17-Node Backbone vs Scoped 2-Hop Subtopology

(Target: 4:3 Network Topology & Subtopology Diagram — Replace in PowerPoint)

Overcoming Semantic Drift: Reverse Prompting & HITL

Two-Layer Semantic Uncertainty (U_{sem})

- Layer 1 (Structural): CFG regex AST check instantly catches syntax errors
- Layer 2 (Semantic): Reverse Prompting reconstructs NL directly from PDDL
- Independent LLM judge measures semantic discrepancy d_{sem} in $[0, 1]$

Uncertainty Formulation:

$$U_{sem} = \begin{cases} 1 & \text{if } v_{struct} = 0 \\ d_{sem} & \text{if } v_{struct} = 1 \end{cases}$$

Fail-Fast HITL Clarification Loop

- Evaluated BEFORE invoking graph solvers or physical tools
- If $U_{sem} > \tau_{sem}$: Pipeline pauses via LangGraph interrupt()
- Prompts operator to clarify ambiguous parameters or missing nodes
- Eliminates infinite trial-and-error conversational loops
- Guarantees that downstream tools receive mathematically verified intent
- If operator approves valid syntax: Bypasses parsing straight to solver

Fail-Fast Guarantee: Zero computational waste on physics simulation when intent is ambiguous

Pre-Deployment Risk Gate: The RADG Decision Function

Piecewise Decision Formulation $D(U_{sem}, QoT_{valid})$:

$$D(U_{sem}, QoT_{valid}) = \begin{cases} \text{clarify} & \text{if } U_{sem} > \tau_{sem} \\ \text{replan} & \text{if } U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 0 \\ \text{approve} & \text{if } U_{sem} \leq \tau_{sem} \wedge QoT_{valid} = 1 \end{cases}$$

🔍 Clarify Intent

Condition: $U_{sem} > \tau_{sem}$

- Trigger: Semantic uncertainty exceeds threshold
- Action: Pause pipeline via interrupt()
- Prompts operator to provide missing data
- Prevents unverified intent from reaching physics

🔄 Suggest Replan

Condition: $U_{sem} \leq \tau_{sem} \wedge QoT = 0$

- Trigger: Valid semantics, but physics infeasible
- Action: Physics failed; notifies operator
- Suggests relaxing constraints (e.g. lower baud rate)
- Loops back to PDDL parsing with feedback

✓ Auto-Approve

Condition: $U_{sem} \leq \tau_{sem} \wedge QoT = 1$

- Trigger: Clear semantics and feasible GSNR
- Action: Autonomous zero-touch provisioning
- Generates auditable planning report
- Zero unverified configurations reach controller

Deterministic Physical Layer: GN-Model QoT Validation

Analytical Gaussian Noise (GN) Engine

- Pure Python implementation: zero LLM arithmetic

Accumulated ASE Noise Power per Span:

$$P_{ASE} = (G - 1) \cdot h \cdot \nu \cdot F \cdot B_{ref}$$

Nonlinear Interference (NLI) Noise Power:

$$P_{NLI} \approx \eta \cdot P_{ch}^3$$

- Accounts for fiber Kerr nonlinearity, dispersion, and EDFA noise

Optical Feasibility & GSNR Criteria

Generalized Signal-to-Noise Ratio (GSNR):

$$GSNR = \frac{P_{ch}}{P_{ASE} + P_{NLI}} \geq GSNR_{th}$$

Receiver Power Sensitivity Budget:

$$P_{rx} = P_{launch} - A_{total} + G_{total} \geq P_{rx,min}$$

- Deterministic feasibility: GSNR >= threshold AND Power >= sensitivity

 Execution Speed: Under 5 milliseconds per candidate route (100% deterministic reproducibility)

Experimental Setup & Testbed Environment

17-Node German Core Network Benchmark

- Realistic telecom benchmark: 17 ROADMs and 26 fiber links
- Standard Single-Mode Fiber (SMF-28): $\alpha = 0.2 \text{ dB/km}$
- Dispersion parameter $D = 16.7 \text{ ps}/(\text{nm}\cdot\text{km})$, $\gamma = 1.2 / (\text{W}\cdot\text{km})$
- Amplified spans: dual-stage EDFAs with noise figure $F = 5.5 \text{ dB}$
- Dynamic link lengths ranging from 45 km to 350 km per span
- Orchestrator: LangGraph StateGraph with memory persistence
- Telemetry: RESTConf and Mock SDON testbed client adapters



Network Topology: Nobel-Germany 17-Node 26-Link Core Backbone

(Target: 16:9 Topology Map with ROADM Nodes & Amplified Spans — Replace in PowerPoint)

Evaluation Framework & Benchmark Scenarios



Architectural Baselines

- Baseline A (LLM-Only): Direct prompt-to-configuration generation with heuristic retry
- Baseline B (Static Rule-Based): Strict regex parser with always-on human review
- Proposed (Neurosymbolic RADG): Decoupled translation + sequential risk gates
- Evaluates LLM reasoning limits vs deterministic safety



100 Test Demands (4 Classes)

- Nominal Intents [40]: Unambiguous valid routing requests with feasible physics
- Ambiguous Intents [20]: Under-specified constraints triggering semantic gate (U_{sem})
- Infeasible Intents [25]: High modulation over long unamplified reaches (QoT fails)
- Adversarial Prompts [15]: Hallucinated nodes & grammar violations



Evaluation Metrics

- Pre-Deployment Blocking Accuracy: % of invalid routes blocked before deployment
- Human Intervention Rate: % of demands requiring human operator clarification
- End-to-End Latency: Pipeline execution time from NL intent to planning report
- Token Consumption: Prompt overhead across scoped vs full topologies

Key Findings & Pre-Deployment Guarantees

100%

Pre-Deployment Safety Guarantee

Zero unfeasible lightpaths reach controller (vs 34% violation in LLM baseline)

> 70%

Reduction in Operator Fatigue

Autonomous approval when $U_{\text{sem}} \leq \tau_{\text{sem}}$ (operator engaged only on high risk)

< 15 ms

Deterministic Physics & Solver Latency

Sub-second orchestration; solver & GN model execute in sub-milliseconds



Empirical Benchmark: GSNR Distribution & Blocking Rate vs Baselines

(Target: 16:9 Matplotlib / Chapter 4 Benchmark Plot — Replace in PowerPoint)

Conclusions & Main Takeaways

Neurosymbolic Decoupling is Essential

- Probabilistic LLMs must never calculate physical impairments or route lightpaths
- Natural language reasoning cleanly bridges to formal PDDL planning
- Guarantees formal soundness without constraining operator expressiveness

Sequential Risk Gates Prevent Failure

- Early semantic evaluation (U_{sem}) eliminates drift before physics computation
- Deterministic GN-model verification (QoT_{valid}) ensures 100% physical feasibility
- Zero unverified configurations ever reach the optical controller

Selective HITL Optimizes Efficiency

- Engaging operators proportionally to risk eliminates fatigue while maintaining safety
- Over 70% reduction in human intervention compared to always-on review
- Sub-second planning latency enables rapid dynamic lightpath turn-up

Production-Ready Engineering

- Fully implemented LangGraph state machine with memory checkpoints
- Benchmarked on realistic 17-node German optical topology and RESTConf testbed
- Establishes a reproducible foundation for autonomous optical networking

Future Outlook & Acknowledgments



Joint Compute and Optical Scheduling

Extending PDDL domain to co-schedule distributed GPU cluster workloads alongside optical lightpaths



Multi-Band & Dynamic Optical Channels

Expanding analytical GN-models to incorporate C+L multi-band SRS inter-channel Raman crosstalk



Autonomous Online Self-Healing

Ingesting real-time testbed telemetry for closed-loop intent re-planning upon physical fiber degradation

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Thank You for Your Attention!

Questions & Discussion